Chapter 9

Quantum key distribution

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Exercises

Problem 9.1: Perfect secrecy

A condition for perfect secrecy in a secure communication scheme is that an eavesdropper can learn nothing about the message given the cipher (the encoded message). That is, the probability for any given message is the same whether or not the cipher is known

$$P(m|c) = P(m) \quad \forall\, m, c. \tag{9.1}$$

(a) Show that this is equivalent to the requirement that every message leads to every possible cipher with equal probability

$$P(c|m) = P(c) \quad \forall\, m, c.$$

Problem 9.2: Six-state BB84

Recall that in the standard version of the BB84 QKD protocol, one party (Alice) chooses randomly between the x- and y-basis and encodes random bits into eigenstates of the Pauli operators $\hat{\sigma}_x$ and $\hat{\sigma}_z$. The states are transmitted to a second party (Bob), who chooses randomly between measuring in the $\hat{\sigma}_x$ or $\hat{\sigma}_z$ eigenbases. A man-in-the-middle (intercept-resend) attack by an eavesdropper (Eve) will introduce an error rate of 25% (so the protocol is secure against intercept-resend attacks only for error rates below 24%).

(a) Suggest a generalisation of the BB84 protocol where Alice uses all six eigenstates of the Pauli operators $\hat{\sigma}_x$, $\hat{\sigma}_y$, and $\hat{\sigma}_z$, and Bob makes measurements of $\hat{\sigma}_x$, $\hat{\sigma}_y$, and $\hat{\sigma}_z$.

(b) Discuss the tolerance of the new protocol to errors and to eavesdropping using intercept-resend attacks. Compare to standard BB84.

Problem 9.3: Coherent-state prepare-and-measure

The *conditional Shannon entropy* of Y given X, for a pair of (classical) random variables X, Y is defined as

$$H(Y|X) = \sum_x p(x) H(Y|X = x) \tag{9.2}$$

$$= -\sum_x p(x) \sum_y p(y|x) \log_2 p(y|x) \tag{9.3}$$

$$= -\sum_{x,y} p(x,y) \log_2 p(y|x). \tag{9.4}$$

It measures the additional information gained upon learning about Y when already knowing X. In particular,

$$H(X,Y) = H(X) + H(Y|X), \tag{9.5}$$

where $H(X,Y)$ is the Shannon entropy of the joint distribution of X and Y.

The *mutual information* is a measure of correlations between random variables. It is defined as

$$I(X:Y) = H(X) + H(Y) - H(X,Y). \tag{9.6}$$

That is, the difference in information content between the joint distribution and the product of the marginals.

(a) Show that $I(X:Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$.

(b) Show that $I(X:Y) = 1$ for two perfectly correlated uniformly random bits and $I(X:Y) = 0$ for two independent uniformly random bits.

Note: For continuous distributions, the sums in the definitions above are simply replaced by integrals and the probabilities by probability densities.

Consider now that one party, Alice, prepares coherent states $|\alpha\rangle$ with α real and following a Gaussian distribution with mean 0 and variance σ^2. Bob performs homodyne detection of the states and measures the $\hat{X}$-quadrature. Let X_A be the random variable associated with Alice's choice of amplitude and X_B the variable associated with Bob's measurement outcome.

(c) Argue that the conditional distribution of Bob's outcome given Alice's preparation is

$$p(x_B|x_A) = \frac{1}{\sqrt{\pi}} e^{-(x_B - x_A)^2}. \tag{9.7}$$

(d) Find the joint distribution $p(x_A, x_B)$.

(e) Compute the mutual information $I(X_A : X_B)$.

In a QKD setting with reverse reconcilliation, Alice and Bob would use error correction in order to correct Alice's data to match Bob's outcomes, i.e. Alice should infer Bob's outcomes. This would be followed by privacy amplification, to reduce correlations with any eavesdropper Eve. They can only achieve secrecy if Alice has stronger correlations with Bob than Eve, i.e. Alice can infer his outcomes better. In the case above, where Alice prepares only states displaced along $\hat{X}$ (only real α), this is not possible.

(f) Suggest an intercept-resend attack which will give Eve (almost) perfect knowledge of Bob's outcome, while keeping the joint distribution $p(x_A, x_B)$ observed by Alice and Bob unchanged. *Hint: assume that Eve can prepare states with any squeezing she likes.*